

Literature Status: Bounded-Gap Quasi-Greediness for Markushevich Bases

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Status. `literature_already_answered`.

Original question. Berasategui and Berná's *Quasi-greedy bases for sequences with gaps* [1], arXiv:2005.07221, proves its bounded-quotient-gap implication under a Schauder-basis assumption. Section 8, Question 1 (page 17 of the 18-page arXiv PDF) asks:

Theorem 5.2 is proven under the condition of Schauder bases. Can the Schauder condition be removed and the result be obtained using only the Markushevich condition?

Identification of the answer. The same authors' later paper *Greedy-like bases for sequences with gaps* [2], arXiv:2009.02257, gives the affirmative answer. Its Theorem 2.11 (page 8 of the arXiv PDF) states:

Let $\mathbf{n}$ be a sequence with bounded quotient gaps. If $\mathcal{B}$ is an $\mathbf{n}$ -quasi-greedy Markushevich basis for a Banach space X , it is quasi-greedy.

This removes exactly the Schauder hypothesis asked about in the source. It also yields the version with a fixed weakness parameter $0 < t \leq 1$: every $\mathbf{n}$ - t -quasi-greedy system is $\mathbf{n}$ -quasi-greedy, since every greedy set is a t -greedy set.

The supporting paper explicitly says that it generalizes the earlier result to Markushevich bases and completes the question. The relation is therefore an acknowledged later-literature answer, not a new result of this run.

Scope. The source's final Question 1 is fully answered. Other questions involving arbitrary quotient gaps, weak-parameter converses, or the later paper's new partial-greediness variants are outside this identification.

References

- [1] M. Berasategui and P. M. Berná, *Quasi-greedy bases for sequences with gaps*, Nonlinear Analysis **208** (2021), 112294; arXiv:2005.07221.
- [2] M. Berasategui and P. M. Berná, *Greedy-like bases for sequences with gaps*, Banach Journal of Mathematical Analysis **18** (2024), article 2; arXiv:2009.02257.